

HERA PERSONAL HEALTH COMPANION

Implementation Assessment & Product Roadmap

Evidence-based review of current capabilities, unimplemented requirements, planned additions, dependencies, and delivery sequence.

Assessment date: 19 September 2026

Scope: Web application, Node.js API, SQLite data layer, and ESP32-S3 wearable

Basis: Runtime source, repository documentation, and *UPDATES.md*

1. Executive Assessment

HERA is a working wellness prototype with wearable telemetry, cycle records and estimates, daily check-ins, water tracking, historical analytics, browser notifications, and an optional Groq-powered assistant. *UPDATES.md* describes a much larger target product. Most target features remain planned and depend on stronger identity, data-quality, and longitudinal-data foundations.

41	21	4	1
advanced target features	sleep requirements	current DB tables	prototype user ID

Scope boundary: HERA must remain informational wellness software. Sleep classification, cycle estimates, anomaly detection, digital-twin outputs, and recommendations must not be represented as diagnosis, treatment, fertility guidance, emergency assessment, or clinically validated prediction.

Status definitions

Status	Meaning
Implemented	Working path exists in runtime code.
Partial / Prototype	Some behavior exists, but requirement is incomplete, local-only, rules-based, or not production-ready.
Planned	Feasible addition with identified foundation and delivery phase.
Future / Research	Needs substantial data, validation, external services, or research before reliable delivery.

2. Current Product Baseline

Capability	Status	Current evidence and limitation
Wearable telemetry	Implemented	ESP32-S3 sends HR, SpO ₂ estimate, movement, activity, wear state, device ID, uptime, and timestamps to REST API.
Cycle tracking	Implemented	Cycle CRUD, duration, average interval, variability, early/on-time/late comparison, predicted range, and provisional phase estimate.
Daily check-ins	Implemented	Mood, stress, energy, perceived sleep, hydration, symptoms, and notes. One record per user/day.
Water and nutrition guidance	Partial / Prototype	Water logging plus rules-based and optional LLM guidance. No food diary, nutrients, image analysis, or personal nutrition baseline.
Analytics	Partial / Prototype	7/30/90-day charts and aggregates. No robust baseline, anomaly, correlation, forecast, or confidence engine.
AI assistant	Partial / Prototype	Bounded recent context, safety prompt, emergency phrase interception, limits, timeout, and rate limiting. Not a model-backed health prediction system.
Profile and notifications	Partial / Prototype	Profile, preferences, reminders, and notification read state are browser-local; no synchronized identity.
PWA	Implemented	Responsive installable shell and service worker; API remains network-only to avoid stale health data.
Accounts and authorization	Not implemented	Several routes use fixed user ID 1. Wearable records are not securely owned by authenticated users.
Production operations	Not implemented	No production deployment, backups, audit logs, monitoring, automated tests, authenticated device ingestion, or compliance review.

3. Advanced Feature Gap Assessment — Personalization and Intelligence

#	Target feature	Status	Roadmap decision
1	Personalized Baseline Learning	Planned	Phase 2. Robust rolling medians/ranges; minimum data threshold; gradual updates; invalid records excluded.
2	Physiological Digital Twin	Future / Research	Start as transparent personal-state profile, not clinical twin. Requires validated baseline and model governance.
3	Personalized Trend Forecasting	Planned	Phase 3. Begin with stable/rising/falling/irregular classification and short horizon.
4	Prediction Stability Monitor	Planned	Phase 3. Persist prior prediction, compare revisions, and reduce confidence on large changes.
5	Symptom–Sensor Discrepancy	Planned	Phase 4 after timestamp alignment and personal baselines. Report discrepancy only.
6	Multi-Parameter Anomaly Detection	Future / Research	Phase 4 prototype. Needs adequate clean longitudinal data and false-alert evaluation.
7	Early Pattern–Deviation Detection	Planned	Phase 3 using rolling windows against personal baseline.
8	Pattern Recurrence Detection	Planned	Phase 4 using normalized feature windows and explainable similarity.
21	Confidence-Aware AI + Explanation	Partial / Prototype	Assistant explains context, but no formal confidence engine. Phase 2 adds deterministic confidence and factor records.
22	Health Data Quality Detector	Partial / Prototype	Basic request validation exists. Phase 1 adds range/noise/gap/duplicate/timestamp checks and exclusion reasons.
23	Sensor-Fusion Intelligence	Planned	Phase 3 after common timestamps, user ownership, and normalized event records.
24	Counterfactual Health Insights	Future / Research	Only after validated model. Must display model simulation and association limitations.
25	Intervention Effectiveness Tracker	Planned	Phase 4 with intervention records and defined before/after windows; no causation claims.
26	Recommendation Ranking	Future / Research	Phase 5 after outcome and preference history exists. Keep rationale and safety rules deterministic.
27	Multimodal Wellness Timeline	Planned	Phase 2. Unified read model over existing and new event tables.
32	AI Health Summary Generator	Planned	Phase 5. Backend selects facts; AI drafts narrative; user edits and approves before export/share.
34	Temporal Pattern Analyzer	Planned	Phase 4 after unified timeline and normalized durations.
35	Cross-Parameter Relationship Mapper	Future / Research	Phase 5. Minimum sample thresholds, association labels, confounder warnings, no causation.

4. Cycle, Activity, Nutrition, and Goals

#	Target feature	Status	Roadmap decision
9	Cycle-Phase Physiological Signature	Future / Research	Requires sufficient phase-tagged sleep, sensor, symptom, and activity history.
10	Cycle-to-Cycle Comparison	Implemented	Current API reports early, late, or on-time against recent interval baseline.
11	Cycle Pattern Change Detection	Partial / Prototype	Multiple intervals feed average/variability, but persistent-change event detection remains Phase 3.
12	Cycle Prediction Confidence	Partial / Prototype	Prediction range reflects variability; explicit confidence and quality note remain Phase 2.
13	Adaptive Cumulative ALI	Planned	Phase 3. Define versioned formula and aggregate activity intervals, not raw sample count alone.
14	Activity Distribution Tracking	Planned	Phase 3. Firmware/backend must preserve durations and sampling gaps.
15	Personal Activity Baseline	Planned	Phase 2/3 after daily activity aggregation.

#	Target feature	Status	Roadmap decision
16–17	Recovery Curve and Change Detection	Future / Research	Need explicit activity bouts, valid post-activity HR series, personalized return-to-baseline model.
18–19	Food Image Processing and Portion Estimation	Future / External	Needs vision API/model, nutrition data source, confidence threshold, and mandatory correction flow.
20	Personalized Nutrition Learning	Planned	Phase 4 after manual food and nutrient records are available.
36	Sleep–Activity Interaction	Planned	Phase 4 after validated sleep and ALI datasets exist.
37	Nutrition–Physiology Association	Future / Research	Phase 5; requires food timestamps, nutrients, sufficient samples, and association-only language.
38	Hydration Pattern Analysis	Partial / Prototype	Daily totals exist. Phase 3 adds longitudinal baseline and change indicators.
39	Rest-Day Balance Tracking	Planned	Phase 4 after reliable daily activity/ALI and rest classification.
40	Wellness Goal Progress	Planned	Phase 3: explicit goals, units, target dates, progress events, and user-controlled reminders.

5. Sleep Monitoring Assessment

All S1–S21 sleep requirements remain unimplemented as wearable sleep monitoring. Current daily check-in stores a subjective 1–5 sleep score only.

Group	Requirements	Status	Delivery assessment
Session and capture	S1–S3	Planned	Firmware sleep mode, timestamps, local buffering, HR and movement sampling.
Sleep estimation	S4–S8	Prototype / Validate	Estimated sleep/wake, onset, wake, duration, interruptions. Validate algorithm; no sleep-stage or disorder claims.
Reports	S9–S10, S14–S15	Planned	Overnight HR profile, component report, dashboard, and aligned timeline.
Personalization	S11–S12	Planned	Needs sufficient complete nights and quality thresholds.
Synchronization	S13	Planned	Idempotent session/chunk IDs, transfer state, retries, and duplicate prevention.
Cross-domain integration	S16–S19, S21	Planned	Only validated values; missing sleep remains unavailable, never zero.
HWS contribution	S20	Future / Research	Blocked until HWS model and sleep contribution receive documented validation and versioning.

6. Healthcare Connection and Accounts

#	Target feature	Status	Dependency / decision
28	Clinic Matching	Future / External	Trusted clinic directory, specialty taxonomy, location search, source freshness.
29	Clinic Profile	Future / External	Display only sourced clinic fields; record source and last verification.
30	Doctor Profile	Future / External	Needs clinic/provider data agreement and update process.
31	Save Specialist	Planned	Requires authenticated profile and per-user persistence.
33	mHealth Specialist Connection	Future / External	User-approved summary plus clinic-supported contact action; no automatic transmission.
41	Multi-Account Profiles	Critical Phase 1	Authentication, authorization, user-owned data, device assignment, sessions, audit, export, deletion.

Privacy gate: No health-summary sharing or clinic contact may happen without explicit user review and approval. No real multi-user health-data deployment before authorization protects every route and record.

7. New Data and Platform Additions

Addition	Purpose	Phase
Users, sessions, and device ownership	Replace fixed user 1; isolate records and authenticate wearable ingestion.	1
Data-quality flags and exclusions	Preserve raw values, validity, reason, source, and processing status.	1
Model/configuration versions	Identify formulas, thresholds, AI model, and configuration used for each result.	1
Derived metrics and insight records	Store reproducible baselines, trends, confidence, factors, and prior predictions.	2
Unified event timeline	Chronological view over sensors, cycles, symptoms, nutrition, sleep, goals, and interventions.	2
Sleep sessions and chunks	Duplicate-safe overnight transfer and processing.	3
Activity bouts and daily ALI	Reliable durations, intensity distribution, and versioned ALI.	3
Food, nutrients, goals, interventions	Enable nutrition learning, progress, and before/after comparisons.	4

Addition	Purpose	Phase
PostgreSQL production persistence	Accounts, concurrency, operations, backups, audit, and mutable production data.	Before production

8. Delivery Roadmap

Phase	Goal	Deliverables	Exit criteria
Phase 0 Truth and tests	Align claims with runtime	Correct implementation-status docs; define statuses; add API/calculation smoke tests; freeze safety language.	Docs match code; core tests repeatably pass.
Phase 1 Trust foundation	Safe identities and clean data	Accounts, authorization, device ownership, quality pipeline, duplicate protection, audit events, model versions, PostgreSQL plan.	No fixed-user access; invalid records excluded with reason; authorization tests pass.
Phase 2 Personalization	Explainable personal reference	HR/activity/cycle/hydration baselines, confidence framework, derived metrics, unified timeline.	Minimum-data rules enforced; calculations reproducible and versioned.
Phase 3 Activity, cycle, sleep MVP	Longitudinal daily intelligence	ALI, activity distribution, cycle confidence/change events, hydration trends, goals, sleep capture/sync/dashboard MVP.	Missing data never zero; sleep transfer idempotent; metrics pass test fixtures.
Phase 4 Cross-domain patterns	Connect records safely	Manual nutrition, interventions, recurrence, discrepancy, temporal analysis, sleep/activity association, rest balance.	Outputs show source period, confidence, and association—not causation.
Phase 5 Advanced/external	Optional intelligent services	Editable AI summary, recommendation ranking, clinic data, specialist saving/contact, food vision, research prototypes.	External sources governed; user approval enforced; validation evidence documented.

9. Recommended Next Release

1. Correct docs/IMPLEMENTATION_STATUS.md, which currently marks REST transmission and backend synchronization as planned despite working runtime paths.
2. Add runnable tests for cycle calculation, validation, wearable ingestion, check-ins, water totals, and authorization boundaries.
3. Replace prototype user ID 1 with authenticated user ownership and device assignment.
4. Add structured data-quality fields/events before building baselines or AI predictions.
5. Deliver personal baseline + confidence + timeline as first new intelligence release.
6. Then implement ALI and sleep capture; defer digital twin, counterfactuals, food vision, and clinic integration.

10. Product Rules That Remain Mandatory

- Informational wellness output only; no diagnosis, treatment, emergency, fertility, or contraception claims.
- Missing values stay unavailable; they are never silently converted to zero.
- AI explains backend-calculated facts; it does not invent measurements or perform hidden clinical scoring.
- Every prediction or insight reports confidence, data sufficiency, key factors, and model/configuration version.
- Food-image output remains editable before save.
- Health summaries remain editable and require explicit approval before sharing.
- Clinic and doctor facts come only from traceable data sources.
- Production launch requires HTTPS, authentication, authorization, secure secrets, backups, monitoring, audit logs, rate limits, retention/export/deletion workflows, and applicable privacy/compliance review.

Prepared from repository state on 19 September 2026. Runtime code is treated as implementation source of truth where planning documents differ.